

1. When area of cross-section of pipe increases its velocity decreases, will be explained by

- A. Archimedes Principle B. Stoke's law
C. Pascal's Law D. Equation of Continuity

Properties of
bulk matter

2. The volume of a block of metal changes by 0.12% when it is heated through 40°C then the coefficient of linear expansion of the metal will be

- A. $6 \times 10^{-5}^{\circ}\text{C}$ B. $1 \times 10^{-5}^{\circ}\text{C}$
C. $2 \times 10^{-5}^{\circ}\text{C}$ D. $3 \times 10^{-5}^{\circ}\text{C}$

3. If the rise in height of liquid in capillary of two tubes is 5.6 cm and 1.4 cm, then the ratio of the radii of the tubes is

- A. 1:2 B. $1:\sqrt{2}$
C. $\sqrt{2}:1$ D. 1:4

4. A Carnot's cycle operating between two temperatures $T_1=500\text{ K}$ and $T_2=300\text{ K}$ produces 1 kJ of mechanical work per cycle. Then the heat transferred to the engine by the reservoir

- A. 2500 J B. 600 J
C. 1800 J D. 1200 J

5. Which of the following is not based upon the law of conservation of energy

- A. Thermodynamics 1st law B. Bernoulli's theorem
C. Kirchhoff's 1st law of electricity D. Lenz' law

Current
electricity

6. Equal volume of all gases contains same number of molecules under the

- A. Constant temperature B. Constant volume
C. Constant pressure D. Similar temperature and pressure

conditions of kinetic
theory of
gases

7. The temperature of the gas is raised from 27°C to 927°C . Then the r.m.s. molecular speed of gas becomes

- A. Half B. Double
C. Remains unchanged D. Four times

8. In a simple harmonic motion, when the displacement is one half the amplitude, then what fraction of the total energy is kinetic

- A. 0 B. $1/2$
C. $1/4$ D. $3/4$

9. A particle in S.H.M. is described by the equation $y=r \cos(\omega t+\theta)$, $\omega=2\pi/T$. If at $t=0$ s position 'y' of the particle is 1 cm and its initial velocity is π cm/s, then its amplitude is

- A. $\sqrt{2}\text{ cm}$ B. 1 cm
C. 0.5 cm D. 1.5 cm

may be
y = 1 cos(2\pi t/T + \theta)

10. A tuning fork of known frequency 256 Hz makes 5 beats per second with a string of an instrument. The beat frequency decreases to 2 beats per second when the tension of the instrument is slightly increased. The frequency of the instrument before increasing the tension was

A. (256+2) Hz ☒ B. (256-2) Hz
☒ C. (256-5) Hz D. (256+5) Hz

11. The identical charges each of $1 \mu\text{C}$ are placed at the corners of an equilateral triangle of side 0.10 m. Then electric field at the center of the triangle will be

A. $27 \times 10^5 \text{ NC}^{-1}$ B. $81 \times 10^5 \text{ NC}^{-1}$
☒ C. Zero D. $3\sqrt{3} \times 10^5 \text{ NC}^{-1}$

12. Total electric flux through any gaussian surface which encloses three dipoles of equal strength, where each dipole has charges $-q$ and $+q$ separated by certain distance 'a' is

A. Zero B. $1/\epsilon_0 \text{ Nm}^2\text{C}^{-1}$
☒ C. $2/\epsilon_0 \text{ Nm}^2\text{C}^{-1}$ D. $6/\epsilon_0 \text{ Nm}^2\text{C}^{-1}$

13. The capacitance required to store energy of 100 kW at a potential difference of 10^4 V is

A. $7.2 \mu\text{F}$ B. 7.2 F
☒ C. 7.2 mF D. $3.6 \mu\text{F}$

14. A 2Ω thick wire is stretched so that its length becomes thrice the earlier one. Assuming there is no change in its density takes place on stretching, then new resistance of the wire becomes

A. 4Ω B. 6Ω C. 9Ω ☒ D. 18Ω
Handwritten notes: $R_2 = (\frac{\text{length}}{\text{radius}})^2 \times R_1$ when doubled new resistance becomes $= 4 R_1$ when tripled new resistance becomes $= 9 R_1 = 9(2) = 18 \Omega$

15. A uniform wire of resistance 60Ω is cut into 5 equal parts and then these pieces are connected in parallel. Then the resistance of the combination is

A. 12Ω B. 2.4Ω C. 3.6Ω D. $1/3 \Omega$
Handwritten notes: $\frac{60 \Omega}{5} = 12 \Omega$
 $\frac{1}{R_p} = \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}$
 $\Rightarrow \frac{1}{R_p} = \frac{5}{12} \Rightarrow R_p = \frac{12}{5}$
 $\Rightarrow R_p = 2.4 \Omega$

16. Energy in a current carrying coil is stored in the form of

A. Electric field only B. Magnetic field only
☒ C. Can be any of the (A) and (B) D. Both (A) and (B)

17. A wire of length L meter carrying a current i ampere is bent in the form of a circle. Then its magnetic moment is

A. iL^2 B. $2\pi L^2 i$
☒ C. $iL^2/4\pi$ D. $iL^2/2\pi$

18. A magnetic field of $2 \times 10^{-2} \text{ T}$ acts at right angles to a coil of area 100 cm^2 with 50 turns. The average emf induced in a coil is 0.1 V , when it is removed from the field in the time t . The value of t is

A. 0.1 s B. 0.01 s
C. 1 s D. 10 s

19. A step up transformer is used on a 120 V line to provide a potential difference of 2400 V . If the primary has 75 turns, then the number of turns in the secondary are $\rightarrow \text{A.C}$

A. 3000 B. 1500
C. 2400 D. 12000

$$\Rightarrow \frac{V_s}{V_p} = \frac{N_s}{N_p} \Rightarrow \frac{2400}{120} = \frac{N_s}{75} \Rightarrow N_s = 1500$$

20. The instantaneous current from a.c. source is given by $V = 300 \sin 240t$. Then root mean square voltage of the source is $\rightarrow \text{A.C}$

A. 191.1 V B. 212.16 V
C. 300 V D. 424.32 V

$$V = V_0 \sin \omega t \Rightarrow V_0 = 300 \sin 240t$$

$$V_{\text{rms}} = \frac{V_0}{\sqrt{2}} = \frac{300}{1.41} = 212.76 \text{ V}$$

21. Biological importance of ozone layer is $\rightarrow \text{E.M. Waves}$

A. It stops ultraviolet rays B. Ozone layer reduces greenhouse effect
C. It reflects radio waves D. It controls C_2/H_2 ratio in the atmosphere

22. One of the following is not an expression of velocity of light, where terms have their usual meanings

A. E_0/B_0 B. $E\lambda/h$
C. $1/\sqrt{(\mu_0 \epsilon_0)}$ D. B_0/E_0

23. Ratio of intensities of two waves are given by $4:1$, then ratio of their amplitudes of the two waves is

A. $2:1$ B. $1:2$
C. $4:1$ D. $1:4$

24. Time taken by sunlight to pass through a window of thickness 4 mm whose refractive index is $3/2$ is

A. $2 \times 10^{-4} \text{ s}$ B. $2 \times 10^8 \text{ s}$
C. $2 \times 10^{-11} \text{ s}$ D. $2 \times 10^{11} \text{ s}$

25. An eye specialist prescribes spectacles having combination of convex lens of focal length 40 cm in contact with a concave lens of focal length 25 cm . The power of this lens combination, in dipotre, is $\rightarrow \text{Optics}$

A. $+1.5$ B. -1.5
C. $+6.67$ D. -6.67

$$P = \frac{100}{f(\text{cm})}$$

$$P = P_1 + P_2 \Rightarrow P = \frac{100}{40} - \frac{100}{25} = 2.5 - 4 = -1.5$$

26. If the K.E. of the free electron doubles its de-Broglie wavelength changes by factor

A. $1/2$ B. 2
C. $1/\sqrt{2}$ D. $\sqrt{2}$

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$\rightarrow$ Dual nature of matter & Radiation

$$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{E_1}{E_2}} = \sqrt{\frac{E_1}{2E_1}} = \frac{1}{\sqrt{2}}$$

27. Which demonstrates the particle nature of the wave → Dual nature of matter & radiation
- A. Hertz experiment B. Plank's quantum theory
- ☒ C. Einstein's photoelectric equation D. Davisson and Germer experiment

28. A nucleus disintegrates into two nuclear parts which have their velocities in the ratio 2:1. The ratio of their nuclear sizes will be

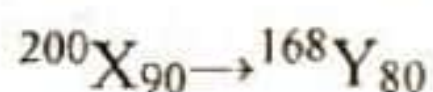
- A. $2^{1/3}:1$ B. $1:3^{1/2}$
- C. $3^{1/2}:1$ ☒ D. $1:2^{1/3}$

29. 1 curie represents

- A. 1 disintegration per second B. 3.7×10^{10} disintegration per second
- C. 3.7×10^7 disintegration per second D. 10^6 disintegration per second

30. The correct arrangement of colours in the descending order of their wave length is → Dual nature of matter & Radiation
- A. Yellow, green, orange, violet B. Yellow, violet, green, orange
- ☒ C. Orange, yellow, green, violet D. Violet, green, yellow, orange
- VIBGYOR

31. What is the respective number of α and β particles emitted in the following radioactive decay?



- A. 8 and 8 B. 8 and 6
- C. 6 and 8 D. 6 and 6
- $^{200}_{90}\text{Xe} \rightarrow ^{168}_{80}\text{Y} + 2^4_2\text{He} + 2^-_1\text{e}$

32. For a transistor $I_c/I_e=0.96$, then current gain for common emitter is

- A. 12 B. 6
- C. 48 ☒ D. 24

33. A piece of copper and another of germanium are cooled from room temperature to 77K, the resistance of → Semiconductors

- A. Each of them increases
- B. Each of them decreases
- ☒ C. Copper decreases and germanium increases
- D. Copper increases and germanium decreases

34. A device in which d.c. power is converted into a.c. power is $\rightarrow$ Semiconductor

- A. A diode
B. An oscillator
C. OR gate
D. Zeener diode

35. The area covered by an antenna of height 100 m is

- A. $4 \times 10^9 \text{ m}^2$
B. $20 \times 10^5 \text{ m}^2$
C. 3600 m^2
D. 360000 m^2

36. In the relation $P = \alpha \beta e^{-\alpha z/kO}$ where α and β are constants, P is pressure, z is distance, k is Boltzmann constant and O is the temperature. The dimensional formula of β is

- A. $[M^0 L^2 T^0]$
B. $[MLT^{-2}]$
C. $[M^1 L^0 T^{-1}]$
D. $[M^0 L^2 T^{-1}]$

37. Find the wrong statement $\rightarrow$ UNITS AND DIMENSIONS

- A. micron = 10^{-6} m
B. 1 fermi = 10^{-15} m
C. 1 barn = 10^{-28} m
D. $1 \text{ A}^\circ = 10^{-8} \text{ cm}$

$1 \text{ A}^\circ = 10^{-10} \text{ cm}$
 $1 \text{ A}^\circ = 10^{-10} \text{ cm}$
 $1 \text{ A}^\circ = 10^{-8} \text{ cm}$

38. An aeroplane moves from station A to the station B with a speed of 1500 km/h and then from station B to the station A with a speed 500 km/h. Then the average speed of the aeroplane is

- A. 1000 km/h
B. 750 km/h ✓
C. 1250 km/h
D. 1200 km/h

$[ML^2 T^{-2} K^{-1}]$

39. A particle is projected at 60° to the horizontal with kinetic energy 160J. The K.E. at the highest point is

- A. 160J
B. 0J ✓
C. 80J
D. 40J

40. Resultant of two equal forces acting at right angles to each other is 282.8N then the magnitude of each force will be

- A. 200N
B. 100N
C. 10N
D. 140N

41. A spring balance is attached to the ceiling of a lift. A man hangs his bag on spring and the spring reads 49N, when the lift is stationary. If the lift moves downward with an acceleration of 5 ms^{-2} , the reading of the spring balance will be

- A. 24N
B. 74N
C. 15N
D. 49N

42. A ship of mass 3×10^7 kg initial is at rest is pulled by a force of 5×10^4 N through a distance of 3m. Assuming that the resistance due to water is negligible, the speed of the ship is

A. 1.5 m/s
 B. 60 m/s
 C. 0.1 m/s
 D. 5 m/s

43. In a rocket fuel burns at the rate of 1kg/s. This fuel is ejected from the rocket with a speed of 60 km/s. This exerts a force on the rocket equal to

A. 6000 N
 B. 60000 N
 C. 60N
 D. 600 N

44. Two masses are in the ratio 1:4 are moving with equal linear momentum. The ratio of their kinetic energies is $\rightarrow$ work energy & power

A. 4:1
 B. 16:1
 C. 1:2
 D. 1:16

$$\frac{m_1}{m_2} = \frac{1}{4} \quad E_1 = \frac{p^2}{2m_1} \quad E_2 = \frac{p^2}{2m_2}$$

$$\frac{E_2}{E_1} = \frac{1}{4}$$

45. A force of 10N is applied on a body for 3s and the corresponding displacement is 6m. The power of the source is

A. 50W
 B. 40W
 C. 30W
 D. 20W

$$F = 10N, s = 6m, t = 3s$$

$$P = \frac{W}{t} = \frac{F \cdot s}{t} = \frac{10 \times 6}{3}$$

$$\Rightarrow P = 20W$$

46. A disc is rotating with angular velocity ω . If heavy mass is placed over it, then what is conserved?

A. Linear momentum
 B. Angular momentum
 C. Kinetic Energy
 D. Moment of inertia

47. If a fly wheel makes 60 revolutions/min, then its angular speed will be

A. 8π rad/s
 B. 4π rad/s
 C. 2π rad/s
 D. π rad/s

48. Value of acceleration due to gravity is maximum at which of the following

A. south pole
 B. center
 C. equator
 D. at certain height

value of g is large at poles

49. If M is the mass of the earth and R is its radius, then the ratio of the gravitational acceleration and the gravitational constant is:

A. M/R^2
 B. R^2/M
 C. M/R^2
 D. $1/MR^2$

$$g = \frac{GM}{R^2}$$

$$\Rightarrow \frac{g}{G} = \frac{M}{R^2}$$

50. When temperature of a gas is 20°C and pressure is changed from

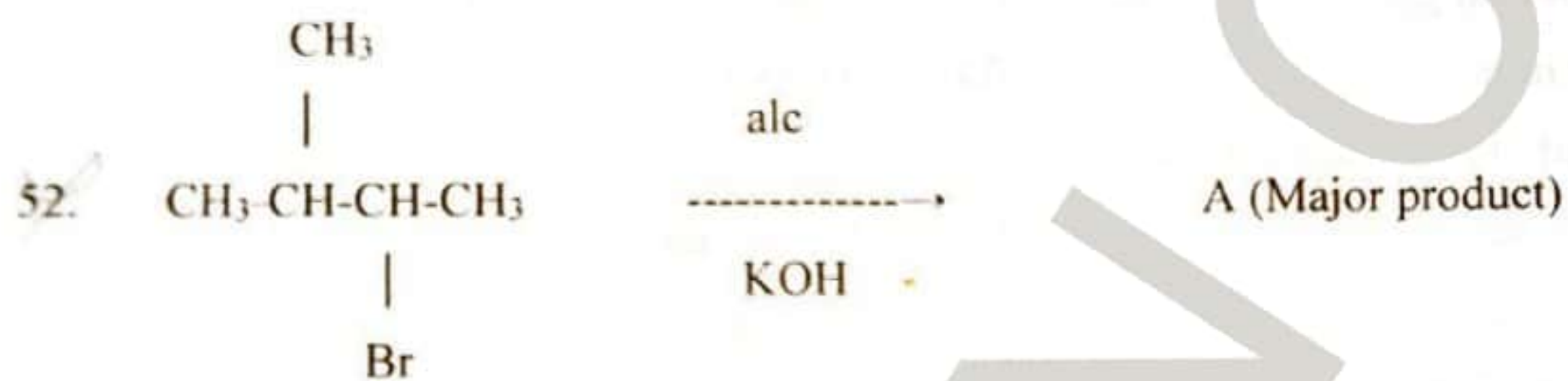
$P_1 = 1.01 \times 10^5$ Pa to $P_2 = 1.165 \times 10^5$ Pa then the volume is changed from 10%. The bulk modulus is

A. 1.55×10^5 Pa
 B. 0.115×10^5 Pa
 C. 1.01×10^5 Pa
 D. 1.01×10^6 Pa

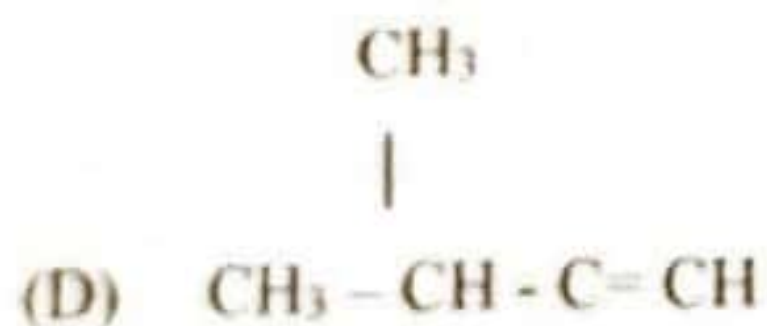
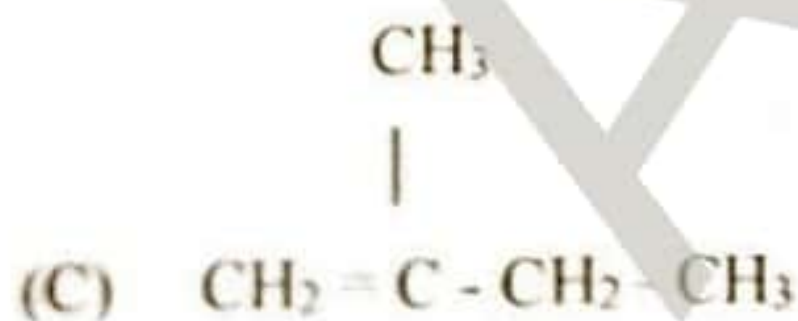
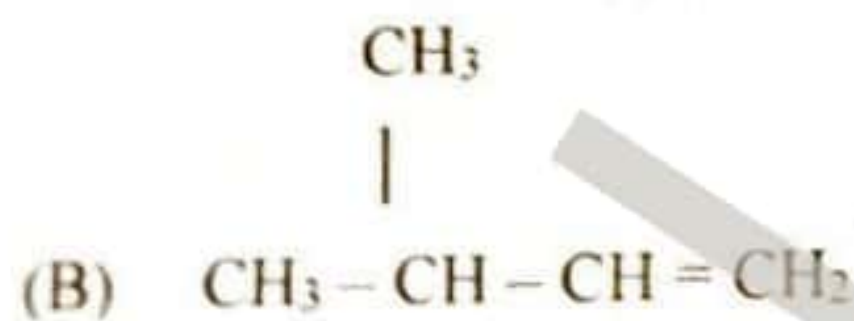
CHEMISTRY

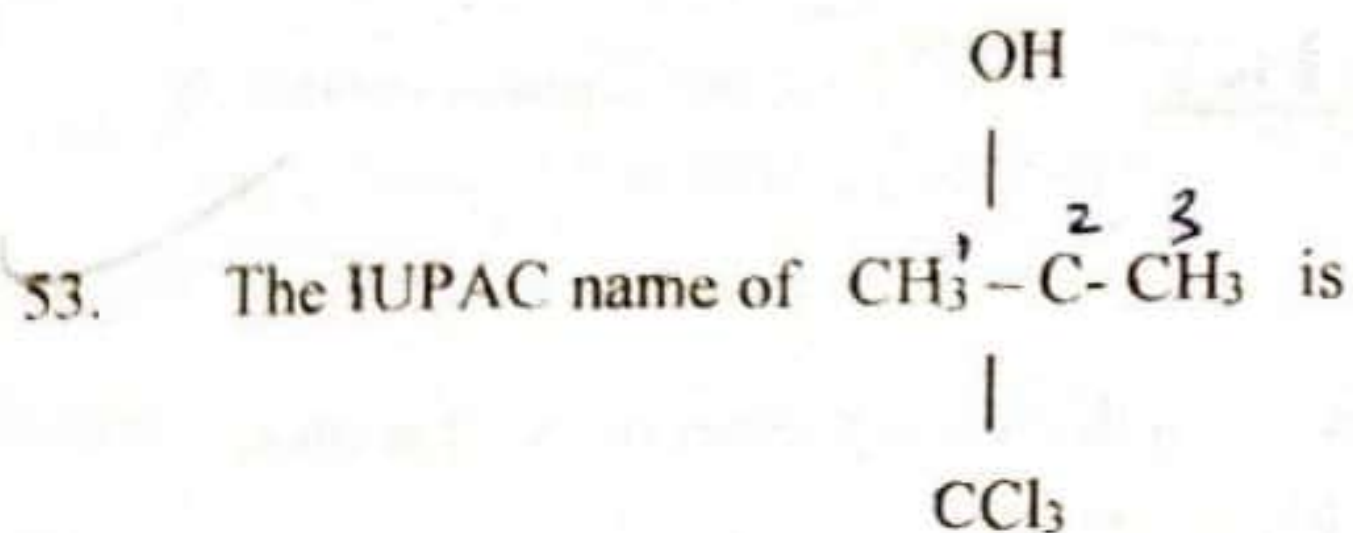
51. Arrange CO_2 ; BF_3 ; BCl_3 ; PH_3 ; H_2S ; SiCl_4 in decreasing order of bond angles.

- (A) $\text{CO}_2 > \text{BF}_3 > \text{BCl}_3 > \text{SiCl}_4 > \text{PH}_3 > \text{H}_2\text{S}$
(B) $\text{CO}_2 > \text{BF}_3 = \text{BCl}_3 > \text{SiCl}_4 > \text{PH}_3 > \text{H}_2\text{S}$
(C) $\text{BF}_3 > \text{BCl}_3 > \text{CO}_2 > \text{PH}_3 > \text{SiCl}_4 > \text{H}_2\text{S}$
(D) $\text{CO}_2 > \text{BCl}_3 > \text{BF}_3 > \text{H}_2\text{S} > \text{PH}_3 > \text{SiCl}_4$



A will be:





- (A) 2-Trichloromethyl-2-propanol
 (B) 2-hydroxy-2-trichloromethylpropane
 (C) 1,1,1-trichloro-2-hydroxy-2-methylpropane
 (D) 1,1,1-trichloro-2-methyl-2-propanol

54. 99% of a first order reaction was completed in 32 min. When 99.9% of the reaction will complete?

- (A) 24 min (B) 8 min
 (C) 4 min (D) 48 min

55. The shape of methyl carbocation is:

- (A) Planar (B) Pyramidal
 (C) Linear (D) V-shaped

56. Oxidation number of carbon in diamond is:

- (A) -4 (B) +4
 (C) 0 (D) +2

57. Which of the following can show optical isomerism?

- (A) $\text{HOCH}_2 - \text{CHCl}_2$ (B) $(\text{CH}_3)_2\text{CHOH}$
 (C) $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$ (D) $\text{ClCH}_2 - \text{CH}_2\text{OH}$

58. Molar heat capacity of ethanol is 110.4 J/K. Its specific heat capacity is:

- (A) 2.4 (B) 55.2
 (C) 5.078 (D) 110.4

59. Heat of neutralization of strong acid against strong base is constant and is equal to:

- (A) -13.7 Kcal (B) -57 KJ
 (C) $-5.7 \times 10^4 \text{ J}$ (D) All the above.

60. The correct order of acidity of N_2O_5 , SO_2 , CO_2

- (A) $\text{N}_2\text{O}_5 > \text{SO}_2 > \text{CO}_2$ (B) $\text{N}_2\text{O}_5 > \text{CO}_2 > \text{SO}_2$
 (C) $\text{SO}_2 > \text{N}_2\text{O}_5 > \text{CO}_2$ (D) $\text{SO}_2 > \text{CO}_2 > \text{N}_2\text{O}_5$

61. The free energy change for a reversible reaction at equilibrium is:

- (A) Zero (B) Positive
 (C) Negative (D) None

62. Which species does not exist?

- (A) SiF_6^{2-} (B) $\text{Sn}(\text{OH})_6^{2-}$
 (C) GeCl_6^{2-} (D) SiCl_6^{2-}

63. Which of the following hydrides of group 15 has maximum basic character?
- (A) NH_3 (B) BiH_3
(C) AsH_3 (D) PH_3
64. Among the Oxo-acids of chlorine, the correct order of increasing oxidising strength is:
- (A) $\text{HClO}_4 < \text{HClO} < \text{HClO}_2 < \text{HClO}_3$
(B) $\text{HClO}_3 < \text{HClO}_2 < \text{HClO}_4 < \text{HClO}$
(C) $\text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2 < \text{HClO}$
(D) $\text{HClO}_4 < \text{HClO}_2 < \text{HClO} < \text{HClO}_3$
65. Which is true for a transition element:
- (A) Its atom in the ground state may have incomplete d-sub shell.
(B) Its ion may have incomplete d-sub shell.
(C) It shows variable oxidation states and its compounds are coloured.
(D) All are correct
66. Which of the following will give colour to an aqueous solution?
- (A) Cu^+ (B) Zn^{2+}
(C) Cr^{3+} (D) Ti^{4+}
67. Acetaldehyde reacts with HCN followed by hydrolysis gives:
- (A) CH_3COOH (B) $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$
(C) $\text{CH}_3\text{CH}(\text{OH})\text{CN}$ (D) $\text{CH}_3\text{CH}_2\text{COOH}$
68. Hoffmann - bromamide reaction is an example of:
- (A) Addition Reaction (B) Substitution Reaction
(C) Elimination Reaction (D) Molecular rearrangement
69. Which of these cannot act as Nucleophilic species?
- (A) NH_3 (B) $\text{NH}_2 - \text{NH}_2$
(C) $\text{NH}_2\text{NH} - \text{Ph}$ (D) Hydroxylamine hydrochloride
70. When phenol is treated with CHCl_3 and NaOH the product formed is:
- (A) Benzaldehyde (B) Salicylaldehyde
(C) Salicylic acid (D) Benzoic acid
71. Adsorption is accompanied by:
- (A) Decrease in Entropy (B) Decrease in Enthalpy
(C) The negative value of $T\Delta S$ (D) All of these
72. Which one of the following will have the highest coagulation power for As_2S_3 colloid?
- (A) PO_4^{3-} (B) SO_4^{2-}
(C) Al^{3+} (D) Na^+
73. Which of the following aqueous solution has least freezing point?
- (A) 1% glucose (B) 1% sucrose
(C) 1% KCl (D) 1% BaCl_2

74. Cellulose is a condensation polymer of:
- (A) Maltose (B) β -glucose
(C) α -glucose (D) β -fructose
75. Polyamide linkage is present in:
- (A) Carbohydrate (B) Lipid
(C) Protein (D) All of the above.
76. PAN is an example of:
- (A) Addition polymer (B) Natural polymer
(C) Condensation polymer (D) All of these.
77. Ethylene diamine is an example of:
- (A) Monodentate ligand (B) Bidentate ligand
(C) Tridentate ligand (D) Hexadentate ligand
78. Co-ordination number and oxidation number of Cr in $K_3[Cr(C_2O_4)_3]$ are respectively:
- (A) 4 & +2 (B) 6 & +3
(C) 3 & +3 (D) 3 & 0
79. Cinnabar is an ore of
- (A) Hg (B) Cu
(C) Pb (D) Zn
80. The hybrid state of Ni in $Ni(CO)_4$ is:
- (A) sp^3 (B) sp^2
(C) sp^3d (D) dsp^2
81. Benzaldehyde reacts with methanol in presence of sodium hydroxide to give benzyl alcohol and sodium methanoate. This reaction is called:
- (A) Cannizaro reaction (B) Aldol condensation
(C) Tischenko reaction (D) Rosenmund reaction
82. Acetylene on ozonolysis produces:
- (A) Glycerol (B) Ethane
(C) Ethanol (D) Dimethylglyoxal
83. Which of the following reducing agents can be used to reduce $RCOOH$ to RCH_2OH :
- (A) $NaBH_4$ (B) $Na/EtOH$
(C) $LiAlH_4$ (D) $H_2/Catalyst$

84. Volume of a mixture of 6.02×10^{23} oxygen atoms and 3.01×10^{23} hydrogen molecules at NTP is:

- (A) 28.0 litre (B) 33.6 litre
(C) 11.2 litre (D) 22.4 litre

85. Gabriel phthalimide synthesis is used in the preparation of:

- (A) Primary amine (B) Secondary amine
(C) Tertiary amine (D) All of these

86. At high pressures, the Vander Waal's equation reduces to

- (A) $PV = RT - a/v$ (B) $PV = RT$
(C) $PV = RT + pb$ (D) $PV = aRT/V^2$

87. The gaseous envelope around the earth is known as atmosphere. The lowest layer of this is extended upto 10 km from sea level. This layer is

- (A) Stratosphere (B) Troposphere
(C) Mesosphere (D) Hydrosphere

88. Which effect causes decrease in density of a crystal?

- (A) Frenkel (B) Schottky
(C) Stoichiometric (D) Electronic

89. Which will liberate H_2 with cold water?

- (A) Fe (B) Al
(C) Cu (D) K

90. The bleaching properties of hydrogen peroxide is due to its:

- (A) Acidic nature (B) Ability to liberate nascent oxygen
(C) Reducing nature (D) Ability to liberate nascent hydrogen.

91. The quantum numbers for the designation of 3d are:

- (A) $n=3, m=-3$
(C) $n=4, l=1$

- (B) $n=3, l=3$

- (D) $n=3, l=2$

(3) $n=3$
 $l = 0 - n - 1$
 $l = 0 - 3 - 1$
 $= 0 - 2$
 $= 0, 1, 2$

3 x 2

92. The order of filling of electrons in the orbitals of an atom is:

- (A) 3d, 4p, 4s, 4d, 5s (B) 4s, 3d, 4p, 5s, 4d
(C) 3d, 4s, 4p, 4d, 5s (D) 5s, 4p, 3d, 4d, 5s

93. Particles in cathode rays have same charge to mass ratio as:

- (A) α -particle (B) β -rays
(C) γ -rays (D) Protons

94. According to given reaction $2\text{Al} + \text{N}_2 \rightarrow 2\text{AlN}$ 27g of Al (Atomic mass of Al = 27) will react with nitrogen equal to:

- (A) 14g (B) 28g
(C) 42g (D) 56g

95. Which has maximum number of lone pairs of electrons on the central atom?

- (A) XeF_2 (B) H_3O^+
(C) XeF_4 (D) XeF_6

96. Total number of σ and π bonds in methyl cyanide [$\text{CH}_3 - \text{CN}$] are:

- (A) 5; 2 (B) 6; 1
(C) 7; 0 (D) 2; 5

97. Which one of the following has the greatest dipole moment?

- (A) CH_4 (B) CCl_4
(C) CH_2Cl_2 (D) CH_3Cl

98. Total number of lone pairs of electrons in XeOF_4 are:

- (A) 1 (B) 3
(C) 12 (D) 15

99. Which of the following is not linear?

- (A) XeF_2 (B) I_3^-
(C) N_3^- (D) SnCl_2

100. Which species has the same shape as NO_3^- ion?

- (A) SO_3 (B) SO_3^{2-}
(C) ClF_3 (D) ClO_3^-

BIOLOGY

101. The main plant body of a bryophyte is
A. Diploid B. Haploid
C. Both (A) and (B) D. None of these
102. Fruit of groundnut (Arachis) is a
A. Nut B. Caryopsis
C. Berry D. Legume
103. Major food crops of the world belong to the family
A. Leguminosae B. Solanaceae
C. Cruciferae D. Graminae
104. A plant cell differs from an animal cell in the absence of
A. Endoplasmic reticulum B. Mitochondria
C. Ribosomes D. Centrioles
105. Seedless vascular plants are known as
A. Bryophytes B. Thallophytes
C. Gymnosperms D. Pteridophytes
106. Stroma is the ground substance of
A. Lysosomes B. Ribosomes
C. Chloroplasts D. Mitochondria
107. Hypanthodium refers to a specialised type of
A. Ovary B. Fruits
C. Inflorescence D. Thalamus
108. Which of the following lists presents a correct order of taxonomic categories?
A. Kingdom, class, phylum, order B. Class, family, order genus
C. Phylum, order, family, species D. Class, order, family, genus *SGIFOC PK*
109. Which of the following is the important absorption region of root?
A. Zone of root hairs B. Root cap
C. Active zone of elongation D. Zone of maturation
110. Starch containing cells become dark violet when treated with
A. Iodine B. Alcohol
C. Acid alcohol D. Glycerin



111. First step in taxonomy is
- A. Description
B. Identification ✓
C. Naming
D. Classification
112. Plectenchyma is a tissue composed of
- A. Aerenchyma
B. Parenchyma
C. Mycelial threads of fungi
D. Tracheids of higher plants
113. Brown colour of skin of earthworm is due to
- A. Melanin
B. Porphyrin
C. Haemoglobin
D. Haemocyanin
114. Pollutant that reduces O₂ carrying capacity of blood is
- A. SO₂
B. CO
C. CO₂
D. NO
115. The term protoplasm was given by
- A. Dujardin
B. R. Brown
C. Fischer
D. Purkinje
116. Prokaryotic genetic system contains
- A. DNA and histone
B. DNA but no histone ✓
C. Neither DNA nor histone
D. Either DNA or histone
117. Nicotinamide can be synthesized in human body by
- A. Tryptophan
B. Arginine
C. Lactose
D. Tyrosine
118. Organization responsible for Red Data Book is
- A. IUCN
B. BNHS
C. ISWL
D. WWF
119. Which is the shortest phase in cell cycle?
- A. G₁
B. G₂
C. S₂
D. M ✓
120. The non-protein part of enzyme is called
- A. Holoenzyme
B. Apoenzyme ✓
C. Prosthetic group
D. None of above

- SERIES-B**

132. Which of the following is controlled by multiple alleles?
- A. Blood groups B. Sickle cell anaemia
C. Colour blindness D. Phenyl ketonuria
133. In protein synthesis the codon used as start signal is
- A. UAA B. CCA
C. AUG D. GCA
134. Humoral immunity is due to
- A. T- Lymphocytes B. B- Lymphocytes
C. L- Lymphocytes D. P- Lymphocytes
135. Metastasis is
- A. Regeneration of cancer cells B. Rapid division of cancer cells
C. Spread of cancer cells D. Slow division of cancer cells
136. ATP is
- A. A nucleotide B. A nucleosome
C. A Purine base D. A nucleoside
137. Pentamerous, actinomorphic flowers, bicarpellary ovary with oblique septa and fruit is a capsule or berry, are characteristic features of
- A. Asteraceae B. Brassicaceae
C. Solanaceae D. Liliaceae
138. In angiosperms, pericycle gives rise to
- A. Primary roots B. Lateral roots
C. Secondary growth D. Cork cells
139. Organisms which obtain their food from non-living material in their environment are
- A. Saprophytes B. Parasites
C. Epiphytes D. Sporophytes
140. During light reaction of photosynthesis, which of the following phenomenon is observed during cyclic phosphorylation as well as non-cyclic phosphorylation?
- A. Formation of ATP B. Formation of NADPH₂
C. Release of O₂ D. Photorespiration
141. Concept of totipotency was given by
- A. Steward B. Haberlandt
C. Guha D. Maheshwari

25
26
134311341124421344343313
142. Part of plant, which is cultured to obtain virus free clones is

- A. Leaf
B. Root tip
C. Shoot tip
D. Embryo

143. The DNA fragment used to detect the targeted DNA is called

- A. DNA chip
B. DNA probe
C. Gel electrophoresis
D. PCR

144. The fruit is a

- A. Ripened ovary
B. Ripened inflorescence
C. Ripened ovule
D. Ripened flower

145. Which of the following ecosystems has the highest gross primary productivity?

- A. Coral reefs
B. Grasslands
C. Rain forests
D. Mangroves

146. Amino acid sequence in protein synthesis is decided by the sequence of

- A. tRNA
B. mRNA
C. cDNA
D. rRNA

147. An environmental agent that triggers transcription from an operon is

- A. Depressor
B. Inducer
C. Regulator
D. Controlling element

148. First life on earth were

- A. Autotrophs
B. Cyanobacteria
C. Photoautotrophs
D. Chemoheterotrophs

149. Binomial nomenclature was introduced by

- A. Linnaeus
B. Darwin
C. Aristotle
D. de Candolle

150. Ustilago and Puccinia are the members of

- A. Deuteromycetes
B. Ascomycetes
C. Basidiomycetes
D. Phycomycetes

MATHEMATICS

151. A parabola has the origin as its focus and the line $x=2$ as its directrix then the vertex of the parabola is at
 (A) (0,1) (B) (2,0) (C) (0,2) (D) (1,0)
152. The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the New set
 (A) remain the same as that of the original set (B) is increased by 2
 (C) is decreased by 2 (D) is 2 times the original median
153. Let $x_1, x_2, x_3, \dots, x_n$ be n observations such that $\sum x_i^2 = 400$, $\sum x_i = 80$, then a possible value of n among the following is
 (A) 9 (B) 12 (C) 15 (D) 18
154. If the system of equations $x+ay=0$, $az+y=0$ and $ax+z=0$ has infinite solutions, then the value of a is
 (A) 0 (B) -1 (C) 1 (D) no real values
155. If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $\det A^3 = 125$, then α is equal to Determinants
 (A) ± 1 (B) ± 2 (C) ± 3 (D) ± 5
Handwritten notes: $\alpha^2 - 4$, $\alpha^2 = 4$, $\alpha = \pm 2$
156. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be such that $f(1) = 3$ and $f'(1) = 6$ then, $\lim_{x \rightarrow 0} \left(\frac{f(1+x)}{f(1)} \right)^{\frac{1}{x}}$ equals
 (A) 1 (B) $e^{1/2}$ (C) e^2 (D) e^3
157. For $x > 0$, $\lim_{x \rightarrow 0} \left((\sin x)^{\frac{1}{x}} + \left(\frac{1}{x} \right)^{\sin x} \right)$ is
 (A) 0 (B) -1 (C) 1 (D) 2
158. If $x^m y^n = (x+y)^{m+n}$, Then dy/dx is
 (A) $x+y/xy$ (B) xy (C) x/y (D) y/x
159. If $f(x) = x^3 + bx^2 + cx + d$ and $0 < b^2 < c$, then in $(-\infty, \infty)$
 (A) $f(x)$ is strictly increasing function (B) $f(x)$ has a local maxima
 (C) $f(x)$ is strictly decreasing function (D) $f(x)$ is bounded
160. A spherical iron ball 10 cm is radius is coated with a layer ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of ice 15 cm, then the rate at which the thickness of ice decreases is
 (A) $\frac{5}{6\pi} \text{ cm/min}$ (B) $\frac{1}{54\pi} \text{ cm/min}$
 (C) $\frac{1}{18\pi} \text{ cm/min}$ (D) $\frac{1}{36\pi} \text{ cm/min}$

161. If the function $f(x) = x^3 + e^{x/2}$ and $g(x) = f^{-1}(x)$ then the value of $g'(1)$ is
 (A) $\frac{1}{2}$ (B) 2 (C) 1 (D) None of these
162. $\int \frac{dx}{x(x^n+1)}$ is equal to
 (A) $\frac{1}{n} \log \left(\frac{x^n}{x^n+1} \right) + C$ (B) $\frac{1}{n} \log \left(\frac{x^n+1}{x^n} \right) + C$
 (C) $\log \left(\frac{x^n}{x^n+1} \right) + C$ (D) $\log \left(\frac{x^n+1}{x^n} \right) + C$
163. $\int \left\{ \frac{(\log x - 1)}{1 + (\log x)^2} \right\}^2 dx$ is equal to
 (A) $\frac{xe^x}{1+x^2} + C$ (B) $\frac{x}{(\log x)^2 + 1} + C$ (C) $\frac{\log x}{(\log x)^2 + C}$ (D) $\frac{x}{1+x^2} + C$
164. $f(x) = \int_{x^2}^{x^2+1} e^{-t^2} dt$ then $f(x)$ is increasing in
 (A) (2,2) (B) no value of x (C) $(0, \infty)$ (D) $(-\infty, 0)$
165. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx$, $a > 0$ is
 (A) 2π (B) $\frac{\pi}{a}$ (C) $\frac{\pi}{2}$ (D) $a\pi$
166. The value of $\int_0^1 \frac{8 \log(1+x)}{1+x^2} dx$ is
 (A) $\frac{\pi}{8} \log 2$ (B) $\frac{\pi}{2} \log 2$ (C) $\log 2$ (D) $\pi \log 2$
167. Area bounded by curves $y^2 = 4x$ and $x^2 = 4y$ is
 (A) 0 (B) $32/3$ (C) $16/3$ (D) $8/3$
168. The solution of the differential equation $ydx + (x+x^2y) dy = 0$ is
 (A) $-\frac{1}{xy} = C$ (B) $-\frac{1}{xy} + \log y = C$
 (C) $\frac{1}{xy} + \log y = C$ (D) $\log y = Cx$
169. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - 4\vec{b}$ are perpendicular to each other, then the angle between $\vec{a}$ and $\vec{b}$ is
 (A) 45° (B) 60° (C) $\cos^{-1}(1/3)$ (D) $\cos^{-1}(2/7)$
170. For what value of 'a' the volume of the parallelepiped formed by the vectors $\hat{i} + a\hat{j} + \hat{k}$, $\hat{j} + a\hat{k}$ and $a\hat{i} + \hat{k}$ is minimum
 (A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $3\sqrt{2}$

171. If the vectors $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} + 4\hat{j} + \hat{k}$ and $\vec{c} = \lambda\hat{i} + \hat{j} + \mu\hat{k}$ are mutually orthogonal, then (λ, μ) is equal to
- (A) $(-3, 2)$ (B) $(2, -3)$ (C) $(-2, 3)$ (D) $(3, -2)$
172. The line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect if k equals
- (A) $3/2$ (B) $9/2$ (C) $-2/9$ (D) $-3/2$
173. Distance between two parallel planes $2x+y+2z=8$ and $4x+2y+4z+5=0$ is
- (A) $7/2$ (B) $5/2$ (C) $3/2$ (D) $9/2$
174. A plane passes through $(1, -2, 1)$ and is perpendicular to two planes $2x-2y+z=0$ and $x-y+2z=4$ then the distance of the plane from the point $(1, 2, 2)$ is
- (A) 0 (B) 1 (C) $\sqrt{2}$ (D) $2\sqrt{2}$
175. A die tossed 5 times getting an odd number is considered a success then the variance of distribution of number of successes is
- (A) $8/3$ (B) $3/8$ (C) $4/5$ (D) $5/4$
176. Two numbers are selected randomly from the set $S = \{1, 2, 3, 4, 5, 6\}$ without replacement one by one the probability that minimum of the two numbers is less than 4 is
- (A) $1/15$ (B) $14/15$ (C) $1/5$ (D) $4/5$
177. If three distinct numbers are chosen randomly from the first 100 natural numbers, then the probability that all three of them are divisible by both 2 and 3 is
- (A) $4/55$ (B) $4/35$ (C) $4/33$ (D) $4/1155$
178. A fair die is rolled the probability that the first time '1' occurs at the even throw, is
- (A) $1/6$ (B) $5/11$ (C) $6/11$ (D) $5/36$
179. It is given that the events A and B are such that $P(A) = 1/4$, $P(A/B) = 1/2$ and $P(B/A) = 2/3$, then $P(B)$ is
- (A) $1/3$ (B) $1/2$ (C) $1/6$ (D) $2/3$

$$P(A \cap B) = P(A) \cdot P(B) - P(A \cup B)$$

$$P(A/B) = P(B)$$

180. The mean and the variance of a binomial distribution are 4 and 2 respectively then probability of two successes is
- (A) $128/256$ (B) $219/56$ (C) $37/256$ (D) $28/256$
181. If $|z| = 1, z \neq \pm 1$, then all the values of $\frac{z}{1-z^2}$ lie on
- (A) a line not passing the origin (B) $|z| = \sqrt{2}$
(C) the x-axis (D) the y-axis
182. Conjugate of a complex number is $\frac{1}{i-1}$, then the complex number is
- (A) $\frac{-1}{i+1}$ (B) $\frac{1}{i-1}$ (C) $\frac{-1}{i-1}$ (D) $\frac{1}{i+1}$
183. The coefficient of the middle term in the expansion in powers of x of $(1+ax)^4$ and $(1-ax)^6$ is same if 'a' equals
- (A) $-3/10$ (B) $10/3$ (C) $-5/3$ (D) $3/5$
184. The numbers of values of x in the interval $[0, 3\pi]$ satisfying the equation $2\sin^2 x + 5\sin x - 3 = 0$ is
- (A) 2 (B) 4 (C) 6 (D) 1
185. The sum of the first n terms of the series $1^2 + 2 \times 2^2 + 3^2 + 2 \times 4^2 + 5^2 + 2 \times 6^2 + \dots$ is $\frac{n(n+1)^2}{2}$ where n is even. When n is odd the sum is
- (A) $\frac{n(n+1)^2}{4}$ (B) $\frac{n^2(n+1)}{2}$ (C) $\frac{3n(n+1)}{2}$ (D) $\left[\frac{n(n+1)}{2}\right]^2$
186. The domain of $\sin^{-1}[\log_3(x/3)]$ is
- (A) $[1, 9]$ (B) $[-1, 9]$ (C) $[-9, 1]$ (D) $[-9, -1]$
187. if $f(x) = \sin x + \cos x$ and $g(x) = x^2 - 1$, then find the interval in which $g(f(x))$ is invertible.
- (A) $[-\pi/4, \pi/4]$ (B) $[-\pi/2, \pi/2]$ (C) $[\pi/4, -\pi/4]$ (D) $[-\pi/4, -\pi/4]$
188. For all complex number Z_1, Z_2 satisfying $|Z_1| = 12$ and $|Z_2 - 3 - 4i| = 5$, the minimum value of $|Z_1 - Z_2|$ is
- (A) 0 (B) 2 (C) 7 (D) 17
189. If Z_1 and Z_2 are two non-zero complex numbers such that $|Z_1 + Z_2| = |Z_1| + |Z_2|$, then $\arg Z_1 - \arg Z_2$ is equal to
- (A) $-\pi/2$ (B) 0 (C) $-\pi$ (D) $\pi/2$
190. The number of real solutions of the equation $x^2 - 3|x| + 2 = 0$ is
- (A) 3 (B) 2 (C) 4 (D) 1

191. The two numbers have A.M. 9 and G.M. 4 then these numbers are the roots of the quadratic equation
- (A) $x^2 + 18x + 16 = 0$ (B) $x^2 - 18x + 16 = 0$
 (C) $x^2 + 18x - 16 = 0$ (D) $x^2 - 18x - 16 = 0$
192. Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If $a < b < c$ and $a + b + c = 3/2$ then the value of a is
- (A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2\sqrt{3}}$ (C) $\frac{1}{2} - \frac{1}{\sqrt{3}}$ (D) $\frac{1}{2} - \frac{1}{\sqrt{2}}$
193. The number of arrangements of the letters of the word BANANA in which the two numbers do not appear adjacently is
- (A) 40 (B) 60 (C) 80 (D) 100
194. The number of seven digit integers, with sum of digits equal to 10 and formed by using the digits 1, 2, 3 only is
- (A) 55 (B) 66 (C) 77 (D) 88
195. $\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \dots + \binom{30}{20}\binom{30}{20}$ is equal to
- (A) $30C_{11}$ (B) $60C_{10}$ (C) $30C_{10}$ (D) $65C_{55}$
196. If the sum of co efficient in the expansion of $(a+b)^n$ is 4096, then the greatest co efficient in the expansion is
- (A) 1594 (B) 792 (C) 924 (D) 2924
197. The sides of a triangle are $\sin \alpha, \cos \alpha$ and $\sqrt{1 + \sin \alpha \cos \alpha}$ for some $0 < \alpha < \frac{\pi}{2}$, then the greatest angle of the triangle is
- (A) 60° (B) 90° (C) 120° (D) 150°
198. If in a triangle ABC $a \cos^2 \left(\frac{C}{2}\right) + c \cos^2 \left(\frac{A}{2}\right)$, then the sides a, b and c
- (A) Satisfy $a+b+c$ (B) are in A.P. (C) are in G.P. (D) are in H.P
199. If the two circles $(x-1)^2 + (y-3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct points then
- (A) $r > 0$ (B) $2 < r < 8$ (C) $r < 2$ (D) $r = 2$
200. If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then C has the value
- (A) 2 (B) -1 (C) 1 (D) -2